

PES UNIVERSITY, BANGALORE
Department of Computer Science and Engineering
SOFTWARE ENGINEERING – LEVEL 3 MINI-PROJECT
DELIVERABLE 1 – SYNOPSIS / PROJECT PROPOSAL

1.1 Project Title and Group

Field	Entry
Project Title	BookNest – Online Bookstore Ordering and Inventory Management System
Group Number	Team 6
Course	UE24CS341A – Software Engineering
Semester	V Semester
Development Methodology	Agile (Scrum, sprint-based)
Technology Stack	JavaScript – MERN Stack (MongoDB, Express.js, React.js, Node.js)
Customer / Stakeholder	Campus Bookstore Manager

1.2 Team Profile and Product Ownership

Member	SRN	Owned functionality	Secondary responsibilities
DHANUSH GOWDA A S	PES1UG24CS812	Customer registration, login and profile management	API review
SAHAJPREET SINGH	PES1UG24CS397	Book catalog, search and filtering	UI/UX
RISHI JAIN	PES1UG24CS375	Cart, checkout and order/payment workflow	Database (MongoDB schema)
S R Lokesh Adithya	PES1UG24CS392	Admin inventory management and sales reporting	Testing / CI-CD

1.3 Problem Statement

A traditional or purely in-person bookstore limits customers to browsing physical shelves, offers no way to check stock or reserve a title before visiting, and gives the store owner no digital record of sales trends or inventory levels. Customers cannot search across genres or authors quickly, cannot track an order once placed, and have no self-service way to review past purchases. BookNest digitizes the ordering journey — cataloguing, search, cart, checkout and order tracking — while giving store administrators a simple way to manage inventory and view operational reports.

1.4 Objectives

- Allow customers to browse and search the book catalog by title, author, genre and price.
- Allow customers to add books to a cart, customize quantity, and place an order with integrated payment.
- Allow customers to track the status of a placed order and view their order history.
- Provide administrators with tools to add, update, and remove books and manage stock levels.

- Provide administrators with basic operational reports (sales, low-stock alerts).
- Develop the product incrementally using Agile practices with sprint-based delivery.

1.5 Functional Features

- Customer registration and login (JWT-based authentication).
- Book catalog browsing, search and filtering.
- Book detail view with description, price, availability and (optionally) ratings/reviews.
- Shopping cart management (add, update quantity, remove).
- Order placement and payment processing (mock/integrated payment gateway).
- Order status tracking and order history.
- Admin catalog and inventory management (add/edit/disable books, stock levels).
- Admin operational reporting (sales summary, low-stock view).

1.6 Short-Term Plan

Iteration	Planned output	Acceptance evidence
Sprint 1	Project setup, requirements baseline, authentication	User stories + Git repo + working login
Sprint 2	Book catalog, search and filtering	Demo + unit tests
Sprint 3	Cart, checkout and order/payment workflow	Integration tests + PRs
Sprint 4	Order tracking and order history	System demo + test evidence
Sprint 5	Admin inventory management and reporting, security hardening	Validation + coverage report
Sprint 6	System validation and release	Final report + deployment/demo

1.7 Synopsis Submission Checklist

- Title, Group Number and Project ID
 - All team members: names and SRNs
 - Problem statement and customer/stakeholder
 - Intended users
 - Objectives and scope
 - Functional features
 - Team ownership matrix
 - Short-term plan
 - Proposed technology/tool stack
 - Agile justification
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